

Shayan Siddiqui

shayansiddiqui.ca | shayansiddiqui172@gmail.com | linkedin.com/in/shayansiddiqui172 | github.com/shayansiddiqui172

Education

Wilfrid Laurier University

Bachelor of Science in Computer Science

Waterloo, Ontario

Expected Graduation: May 2027

Experience

Software Engineer Intern

Jan 2026 – Present

Wealth Capital Connections

Ottawa, Ontario

- Engineered a **real-time data ingestion pipeline** using **Python** and **REST APIs** to aggregate engagement metrics, replacing a manual research process and saving the team **2 hours per week**
- Automated trend analysis by building a scheduled **ETL workflow** using pandas and SQL to identify recurring patterns, improving front-end iteration speed by **20%**
- Deployed a **live-updating visualization tool** on a Firebase real-time database, increasing user retention by **5%**

Software Analyst

Mar 2025 – Dec 2025

Avis Car Rental

Toronto, Ontario

- Extended internal command-line tooling using **Bash** and **Python** to automate fleet inventory operations, reducing system latency by **2.5%** across multi-location infrastructure
- Optimized **SQL queries and data transformation pipelines** for fleet logistics, improving data accuracy by **15%**
- Digitized and standardized vehicle tracking records through structured automation scripts, accelerating the rental check-in workflow

Software Engineer Intern

Apr 2024 – Sep 2024

Habitat For Humanity

Milton, Ontario

- Developed front-end features using HTML, CSS, and JavaScript to redesign the navigation structure, increasing website engagement by **3%**
- Implemented a **version-controlled document management system** using Git and GitHub, saving **1.5 hours** of administrative labor per week

Projects

HTTP/1.1 Server & Private Cloud | Systems Project

Go, TCP/IP, Linux

- Architected an **HTTP/1.1 server from scratch** in Go using raw TCP, implementing request parsing, concurrent connection handling, and routing without any external frameworks
- Deployed as a **self-hosted private cloud** with streaming file I/O on Linux, supporting multi-gigabyte transfers at constant memory overhead

CartSniper | Full Stack Platform | HackCanada 2026

TypeScript, Node.js, Firebase, REST APIs

- Designed and built a **backend ingestion pipeline** that processes live grocery flyer data for **1,000+ items** in under **1 second**, integrating barcode scanning and image recognition via REST API calls
- Architected a Firebase-backed real-time data layer that surfaces price comparisons across 10+ Canadian retailers, reducing average cart cost by **23%**

Tab Organizer | Production Chrome Extension | Featured by Google | 600+ Installs

JavaScript, Chrome APIs

- Built a tab classification algorithm using Chrome APIs to auto-group 100+ tabs with 95%+ accuracy, deployed to production and **featured by Google** with **600+ active installs**
- Implemented session persistence and crash recovery via chrome.storage API, supporting 50+ saved browsing sessions

Transformer LLM | AI & Systems

Python, PyTorch, TensorFlow

- Implemented a **transformer-based language model from scratch** in Python, covering tokenization, attention mechanisms, positional encoding, and a custom training loop using PyTorch

Technical Skills

Languages: Python, Go, JavaScript, TypeScript, SQL, Java, Bash, C

Frameworks & Libraries: FastAPI, Node.js, Express.js, REST APIs, PyTorch, TensorFlow, pandas, Apache Kafka

Database & Cloud: PostgreSQL, Firebase, Docker, Kubernetes, AWS, Azure

Developer Tools: Git, GitHub, CI/CD, Linux, Jira, VS Code, Postman, Vim, Claude Code (MCP, Skills)